



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2025-26

Class:	BE	Semester:	VIII
Course Code:		Course Name:	PM

Name of Student:	BART ANKIT VINOD
Roll No. :	61
Assignment No.:	2
Title of Assignment:	
Date of Submission:	
Date of Correction:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Completeness	5	4
Demonstrated Knowledge	3	3
Legibility	2	2
Total	10	9

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Completeness	5	3-4	1-2
Demonstrated Knowledge	3	2	1
Legibility	2	1	0

Checked by

Name of Faculty : Dr. Tadwadgahe RN

Signature :

Date :


9/12

Q. 1) A final-year engineering group has been assigned a capstone project to develop an IoT-based smart irrigation system. In the first few weeks, team members are polite but unsure about their roles. Later disagreements arise regarding technical design choices. After some time, they successfully complete the project and present it confidently. Question - Identify and explain the stages of team development depicted in this scenario. How did each stage contribute to the successful completion of the project?

→ Team development model -

forming → storming → norming → performing → adjourning

- forming stage
 - Members are cautious and polite
 - roles and responsibility unclear
 - dependence on leader/instructor
 - limited conflict
- storming stage
 - conflict and disagreements
 - difference in opinions
 - competition for ideas
 - possible frustration
- norming stage
 - Rules become clear
 - cooperation increases
 - trust develops
 - conflicts are resolved constructively
- performing stage
 - high efficiency
 - strong collaboration
 - minimal supervision needed
 - focus on results
- adjourning stage
 - Team disbands
 - reflection on achievements
 - recognition of contributions

Stage	Team behavior	Contribution to
forming	Polite, uncertain	goal clarity & initial
storming	conflict & debate	better technical decision
norming	cooperation	improved coordination
performing	high productivity	successful completion
adjourning	closure	learning & recognition

The scenario reflects Tuckman's stages of team development: forming, storming, norming and performing, each played a critical role in the team's success. forming establishes initial relationships, storming improved decisions quality through constructive conflict, norming enhanced collaboration and task clarity, and performing led to efficient project execution and confident presentation. Together, these stages enabled the team to successfully complete this IoT-based smart irrigation system project.

2) Calculate the net present value for a project that costs Rs. 2,40,000/- and would yield after tax as follows. Assumes cost of capital is 10%. Comment on feasibility of project based on NPV.

→ Initial investment = 2,40,000

Year	Cash flow
1	25,000
2	75,000
3	80,000
4	1,00,000

S1- Calculate present values -

- discount factors at 10%.

$$\text{Year 1} \rightarrow 1 / (1.1) = 0.909 \times 25000 = 22,725$$

$$\text{Year 2} \rightarrow 1 / (1.1^2) = 0.826 \times 75000 = 61,950$$

$$\text{Year 3} \rightarrow 1 / (1.1^3) = 0.751 \times 80000 = 60,080$$

$$\text{Year 4} \rightarrow 1 / (1.1^4) = 0.683 \times 100000 = 68,300$$

∴ Total 2,13,055

S2- Calculate NPV -

$$\text{NPV} = 2,13,055 - 2,40,000$$

$$= -26,945$$

Decision -

- NPV is negative

Thus - Project is not feasible

- Reject the project

because NPV is negative (-26,945)

Q. 3) Consider a project with the following cash flow system. The cost of capital (c) for the firm is 10%. calculate NPV of project and decide whether to accept or reject the project?

→ given, year	cash flow
0	-10,00,000
1	2,00,000
2	2,00,000
3	3,00,000
4	3,00,000

S1 - Present values -

discount factors same as above -

$$\begin{aligned}
 \therefore \text{Year 1} &\rightarrow 2,00,000 \times 0.909 = 1,81,800 \\
 \text{Year 2} &\rightarrow 2,00,000 \times 0.826 = 1,65,200 \\
 \text{Year 3} &\rightarrow 3,00,000 \times 0.721 = 2,25,300 \\
 \text{Year 4} &\rightarrow 3,00,000 \times 0.683 = 2,04,900
 \end{aligned}$$

$\therefore$ Total PV of inflows 1,81,200

$$\begin{aligned}
 &= 1,65,200 + 2,25,300 + 2,04,900 = \\
 &= 7,77,200
 \end{aligned}$$

S2 - NPV -

$$\begin{aligned}
 \text{NPV} &= 7,77,200 - 10,00,000 \\
 &= -2,22,800
 \end{aligned}$$

- decision - NPV is negative (-2,22,800)
- project should be rejected.

because, NPV is negative which is -2,22,800

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